



Barun Shahi



Meenakshi Kumari

infected area within a second.

Tested and recommended for use in hospitals by India's top medical bodies, including AIIMS, ICMR, and IMA, the full-body disinfectant machine was awarded the IMA National President Award in 2021. Shahi has filed for a patent for the machine, which is currently in the last stage of the grant.

The startup is engaging in a discussion with the state health department to deploy this machine in state healthcare facilities and hospitals. The company intends to commercialise the product after the patent grant is received.



Nikky Kumar Jha

A HOME FOR VEGGIES

Necessity is surely the mother of invention, and that is what prompted Nikky Kumar Jha, an alumnus of Nalanda University in Bihar, to come up with the idea of a green storage system for vegetables and fruits for farmers and retailers.

After watching farmers and vegetable vendors in his village Naya Tola Dudhela in Bhagalpur, Bihar, waste a large quantity of their produce due to lack of adequate storage facilities and electricity, Jha started working on a storage system in 2019 to increase the shelf life of fruits and vegetables.

Jha set up Saptkrishi Scientific Pvt Ltd and came up with a solar-powered storage device that requires a power supply of just 10 watts and one litre of water every day, whose production started in March 2021.

The device consists of a high-humid and sterile isolated chamber made of nano-surface reflective coating material. An ARM based embedded system device with a programmable logic controller allows the user to select different modes of operation for different vegetables and fruits. It oxidises ethylene emitted by the produce into small molecules, thus delaying browning and ripening, and regulates the ac-

tivity of the antioxidant enzyme. This creates a controlled microclimate that preserves fruits and vegetables for anywhere between 3 to 30 days.

Started with an initial seed amount of ₹1.5 million, the startup has set up a manufacturing facility that now makes 100 units per month. Each unit can accommodate 200kg of produce and measures approximately 122cm × 107cm × 76cm (4 feet × 3.5 feet × 2.5 feet). Priced at ₹10,000 per unit, the startup is collaborating with different microfinance companies to assist its target customers, which include the local vegetable vendors and farmers, get the product on an EMI basis.

The company plans to collaborate with different state governments to help the farmers get the product at a subsidised rate. The company has sold more than 500 units in different parts of the country till now and has an order of 750 units in its pipeline.

Aiding innovators from all walks of life, incubation centres like IIT Patna aren't just looking at degrees and backgrounds but assessing ideas and supporting them to develop products. What remains to be seen is how these products effectuate the purpose of their invention and become a part of our daily lives. **EFY**

—Yashasvini Razdan